

Introduction to Economics

Chapter 1: Basics of Economics

Learning intentions

- Define 'Economics'.
- Explain the fundamental economic problem of **scarcity** (because resources are scarce while people's wants (not needs) are unlimited).,
- Explain the need for individuals, firms and governments to make **choices** (due to scarcity of resources).
- Define **opportunity cost** and explain how it results from the need to make choices.
- Explain the basic questions of **resource allocation**.

Introduction

Have you ever heard anything about Economics?

Introduction

- Have you ever heard anything about Economics?
- It is obvious you heard about economics and even you talked a lot about economics in your day to day activities.
- you may have questions such as:
 - What are resources?
 - What does efficient allocation mean?
 - What are human needs?
 - What does demand mean?
 - What is economics?
 - etc...

Introduction

- This course will answer those questions and introduce you to
 - the nature of economics,
 - demand and supply theories,
 - theory of consumer,
 - theory of production,
 - theory of cost,
 - theory of market structure and
 - fundamental concepts of macroeconomics at large.

Chapter objectives

- After successful completion of this chapter, you will be able to:
 - understand the concept and nature of economics;
 - analyze how resources are efficiently used in producing output;
 - identify the different methods of economic analysis ;
 - distinguish and appreciate the different economic systems;
 - understand the basic economic problems and how they can be solved; and
 - identify the different decision making units and how they interact with each other.

1.1 Definition of economics

- Economics is one of the most exciting disciplines in social sciences.
- The word **economy** comes from the **Greek** phrase “one who manages a household”.
- The science of economics in its current form is about two hundred years old.
- **Adam Smith** – generally known as the father of economics – brought out his famous book, “***An Inquiry into the Nature and Causes of Wealth of Nations***” , in the year 1776.
- Though many other writers expressed important economic ideas before Adam Smith, economics as a distinct subject started with his book.

1.1 Definition of economics

- There is no universally accepted definition of economics. This is because different economists defined economics from different perspectives:
 - a. Wealth definition,
 - b. Welfare definition,
 - c. Scarcity definition, and
 - d. Growth definition
- Hence, its definition varies as the nature and scope of the subject grow over time. But, the formal and commonly accepted definition is as follow.

1.1 Definition of economics

- Economics is a social science which studies about efficient allocation of scarce resources so as to attain the maximum fulfillment of unlimited human wants.
- As economics is a science of choice, it studies how people choose to use scarce or limited productive resources (land, labor, equipment, technical knowledge and the like) to produce various commodities.

1.1 Definition of economics

- The following statements are derived from the above definition.
 - Economics studies about scarce resources;
 - It studies about allocation of resources;
 - Allocation should be efficient;
 - Human wants are unlimited
 - The aim of economics is to study how to satisfy the unlimited human wants up to the maximum possible degree by allocating the resources efficiently.

1.1 Definition of economics

- The ‘social’ aspect is because economics looks at human behavior, particularly in relation to satisfying human needs and wants.
- Economics is also a ‘science’. This is because of the way that economists put forward and investigate theories in the same way as scientists.

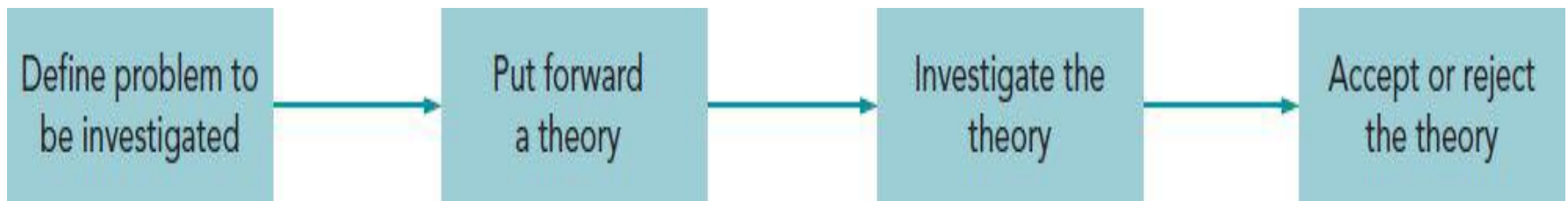


Figure 1.1: Economics as a science

1.2 The rationales of economics

- There are **two fundamental facts** that provide the foundation for the field of economics.
 1. Human (society's) material wants are unlimited.
 2. Economic resources are limited (scarce).
- The **basic economic problem** is about **scarcity** and **choice** since there are only limited amount of resources available to produce the unlimited amount of goods and services we desire.

1.2 The rationales of economics

- Thus, economics is the study of how human beings make **choices** to use scarce resources as they seek to satisfy their unlimited wants. Therefore, choice is at the heart of all **decision-making**.
- As an individual, family, and nation, we confront difficult choices about how to use limited resources to meet our needs and wants.

1.2 The rationales of economics

■ Economists

- study how these choices are made in various settings;
- evaluate the outcomes in terms of criteria such as efficiency, equity, and stability; and
- search for alternative forms of economic organization that might produce higher living standards or a more desirable distribution of material well-being.

1.3.1 Scope of Economics

- The field and scope of economics is expanding rapidly and has come to include a vast range of topics and issues.
- In the recent past, many new branches of the subject have developed, including development economics, industrial economics, transport economics, welfare economics, environmental economics, and so on.
- However, the core of modern economics is formed by its two major branches: **microeconomics** and **macroeconomics**.

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Branches of Economics

Microeconomics

- a branch of economics that is concerned with the analysis of **individual parts of the economy**.
- looks into economic behavior of individual decision making units such as households, firms, markets and industries.

Eg. factors determining the behaviour of a consumer/a firm, the demand for/the supply of a good, the price of a good, the quantity of a good, the performance of a market, etc.

Macroeconomics

- a branch of economics that is concerned with the analysis of the **whole economy**/looks at the big picture of the economy.
- deals with the effects and consequences of the aggregate behavior of all decision making units in a certain economy.

Eg. factors determining aggregate variables such as aggregate demand, aggregate supply, national output, unemployment, inflation, the balance of payments, etc.

1.3.2 Positive versus Normative Analysis

Positive economics

- concerned with analysis of facts and is verifiable. It doesn't have to be necessarily true.
- attempts to describe the world as it is and answer the questions **what was; what is; what will be?**
- Disagreement over positive economics can be settled by an appeal to **facts**.

Normative economics

- A matter of opinion / value judgments. It can be true or false, but can not be tested.
- It tells us **what should be/what ought to be?**
- It evaluates the desirability of alternative outcomes based on one's value judgments about what is good or what is bad.
- Deciding what is good or bad policy is not just a matter of science. It also involves our views on ethics, religion, and political philosophy.

1.3.2 Positive versus Normative Analysis

Examples

- *“Inflation is rising in Ethiopia.”*
- *“Welfare programs reduce the incentive for people to work”*
- *Raising taxes on the wealthy slows economic growth.*
- *“A decrease in personal income tax will lead to a rise in unemployment.”*
- *“Paying people who do not work is wrong and unfair.”*
- *The government **should** raise taxes on the wealthy to pay for helping the poor.”*
- *“A redistribution of income from the rich to the poor will increase social welfare.”*

The above statement is a positive statement as it is verifiable (both personal income tax and unemployment are measurable)

It is a normative statement as it is not verifiable (Though redistribution of income is measurable, social welfare is not).

1.3.2 Positive versus Normative Analysis

Exercise: Write 5 Positive statements and 5 Normative Statements

1.3.3 Inductive and Deductive reasoning in economics

- The fundamental objective of economics, like any science, is the establishment of valid generalizations about certain aspects of human behavior.
- Those generalizations are known as theories.
- A theory is a simplified picture of reality.
- Economic theory provides the basis for economic analysis which uses logical reasoning. There are two methods of logical reasoning: **inductive** and **deductive**.

1.3.3 Inductive and Deductive reasoning in economics

An economic theory derives laws/generalizations through two methods:

1. Inductive Method/Empirical method (bottom-up approach)

- Logical method of reaching at a correct general statement or theory based on several independent and specific correct statements. / deriving a principle or theory by moving from facts to theories and from particular to general economic analysis.

Eg. Formulation of the generalization of diminishing returns.

The Malthusian theory of population - human population grows more rapidly than the food supply until famines, war or disease reduces the population.

1.3.3 Inductive and Deductive reasoning in economics

- **Inductive method involves the following steps.**
 1. Selecting problem for analysis
 2. Collection, classification, and analysis of data
 3. Establishing cause and effect relationship between economic phenomena.

1.3.3 Inductive and Deductive reasoning in economics

2. Deductive Method (top-down approach)

- Logical way of arriving at a particular/specific correct statement starting from a correct general statement.
- It deals with conclusions about economic phenomenon from certain fundamental assumptions or truths or axioms through a process of logical arguments.

Eg. any supply and demand analysis you do is the application of generally accepted principles about demand and about supply

❖ Both methods have weaknesses, so we cannot rely exclusively in one method. They are complementary.

1.3.3 Inductive and Deductive reasoning in economics

- **Major steps in the deductive approach include:**
 1. Problem identification
 2. Specification of the assumptions
 3. Formulating hypotheses
 4. Testing the validity of the hypotheses

1.4 Scarcity, Choice, Opportunity cost and Production Possibilities Frontier(PPF)

1. Have you ever faced a problem of choice among different alternatives? If yes, what was your decision?
2. What is scarcity? Do you think that it is different from shortage? Why?

1.4.1 Scarcity

- The fundamental economic problem that any human society faces is the problem of **scarcity**.
- Scarcity refers to the fact that all economic resources that a society needs to produce goods and services are **finite** or **limited** in supply.
- But their being limited should be expressed in relation to **human wants**.
- Thus, the term scarcity reflects the **imbalance** between our wants and the means to satisfy those wants.

1.4.1 Scarcity

- It is the excess of wants resulting from having limited resources (land, labor, capital and entrepreneurs) in satisfying the endless wants of people.
- It is a **universal problem** for societies – it is not limited to poor countries.
- To the economist, all goods and services that have a **price** are relatively **scarce**. This means that they are scarce relative to people's demand for them.

Resources

1. **Free resources:** A resource is said to be **free** if the amount available to a society is **greater** than the amount **people desire** at **zero price**. E.g. sunshine
2. **Scarce (economic) resources:** A resource is said to be **scarce** or economic resource when the amount available to a society is **less** than what **people want** to have at **zero price**.

Examples of scarce resources.

- All types of human resources: manual, intellectual, skilled and specialized labor;
- Most natural resources like land (especially, fertile land), minerals, clean water, forests and wild - animals;
- All types of capital resources (like machines, intermediate goods, infrastructure); and
- All types of entrepreneurial resources.

Economic resources

- Economic resources are usually **classified** into **four** categories.

- 1. Labor:** refers to the physical as well as mental efforts of human beings used in the production and distribution of goods and services. The reward for labor is called **wage**.
- 2. Land:** refers to the natural resources or all the free gifts of nature usable in the production of goods and services. The reward for the services of land is known as **rent**.

Economic resources

- 3. Capital:** refers to all the manufactured inputs that can be used to produce other goods and services.
- All non-natural (manufactured) resources that are used in the creation and production of other products
 - **Example:** equipment, machinery, transport and communication facilities, etc.
 - The reward for the services of capital is called **interest**.

Economic resources

4. Entrepreneurship:

- Refers a special type of human talent that helps to organize and manage other factors of production to produce goods and services and takes risk of making losses.
- The reward for entrepreneurship is called **profit**.

Note:

Scarcity is the fundamental economic problem. It is natural phenomenon that refers to a state when a resource is available in a finite quantity. It is **permanent**.

Shortage is a market phenomenon (which occurs whenever quantity demanded is greater than quantity supplied at the market price) and is **temporary**.

Economic resources

- **Entrepreneurs are individuals who:**
 - **Organize factors of production to produce goods and services.**
 - **Make basic business policy decisions.**
 - **Introduce new inventions and technologies into business practice.**
 - **Look for new business opportunities.**
 - **Take risks of making losses.**

Factors of Production

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graph TD; A[Factors of Production] --> B((Land)); A --> C((Labor)); A --> D((Capital)); A --> E((Enterprise)); B --> F[Rent]; C --> G[Wages]; D --> H[Interest]; E --> I[Profit]; F --> J[INCOME]; G --> J; H --> J; I --> J;
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The diagram is a flowchart illustrating the flow from factors of production to income. At the top is a yellow rectangular box labeled 'Factors of Production'. Four arrows point downwards from this box to four colored circles: a light blue circle for 'Land', a light green circle for 'Labor', a light orange circle for 'Capital', and a light purple circle for 'Enterprise'. From each of these circles, an arrow points down to a corresponding colored square: 'Rent' (light blue), 'Wages' (light green), 'Interest' (light orange), and 'Profit' (light purple). Finally, four arrows point from these four squares to a single yellow rectangular box at the bottom labeled 'INCOME'. The text 'Payments to factors of Production' is written vertically on the left side of the diagram.

Land

Labor

Capital

Enterprise

Payments
to factors
of
Production

Rent

Wages

Interest

Profit

INCOME

1.4.2. Choice

- All economic questions and problems arise from **scarcity**.
- Economics assumes people **do not** have the resources to satisfy all of their wants.
- Therefore, we must make **choices** about how to allocate those resources. We make decisions about how to spend our money and use our time.

1.4.2. Choice

- Scarcity → limited resource → limited output → we might not satisfy all our wants → make choice → choice involves costs
- In short, scarcity implies choice. Choice, in turn, implies cost.

1.4.3. Opportunity cost

- In a world of **scarcity**, a decision to have **more** of one thing, at the same time, means a decision to have **less** of another thing.
- Decisions involve tradeoffs. When you make a choice, you give up an opportunity to do something else.
- The **value** of the **next best alternative** that must be sacrificed is, therefore, the opportunity cost of the decision.

1.4.3. Opportunity cost

- In a world of **scarcity**, a decision to have **more** of one thing, at the same time, means a decision to have **less** of another thing.
- The **value** of the **next best alternative** that must be sacrificed is, therefore, the opportunity cost of the decision.
- **Definition:** Opportunity cost is the amount or value of the next best alternative that must be sacrificed (forgone) in order to obtain one more unit of a product.

1.4.4. The Production Possibilities Frontier or Curve (PPF/ PPC)

- **PPF/PPC:** is a graph that shows the combinations of output that the economy can possibly produce given the available factors of production and the available production technology

1.4.4. The Production Possibilities Frontier or Curve (PPF/ PPC)

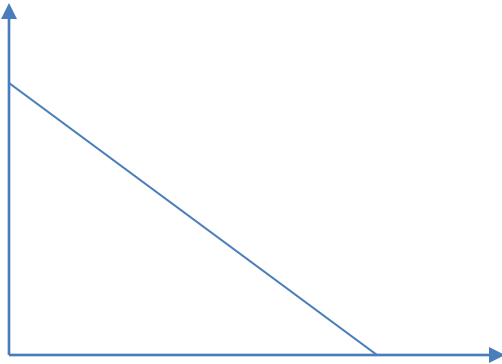
- Consider an economy that produces only **two goods**—**cars** and **computers**.
- Together, the car industry and the computer industry use all of the economy's factors of production.
- The production possibilities frontier is a graph that shows the various combinations of output—in this case, cars and computers—that the economy can possibly produce given the available factors of production and the available production technology that firms use to turn these factors into output.

1.4.4. The Production Possibilities Frontier or Curve (PPF/ PPC)

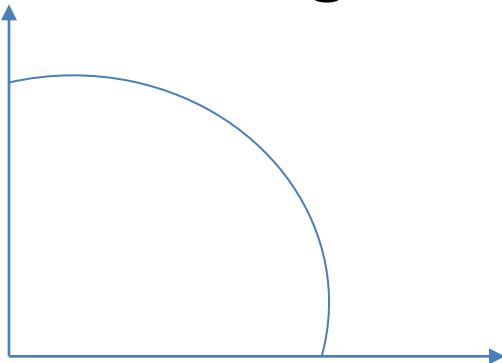
- To draw the PPF we need the following assumptions.
 1. The quantity as well as quality of economic resource available for use during the year is fixed.
 2. There are two broad classes of output to be produced over the year.
 3. The economy is operating at full employment and is achieving full production (efficiency).
 4. Technology does not change during the year.
 5. Some inputs are better adapted to the production of one good than to the production of the other (specialization).

Examples

Eg. 1. constant opportunity cost



Eg. 2. Increasing marginal opportunity cost



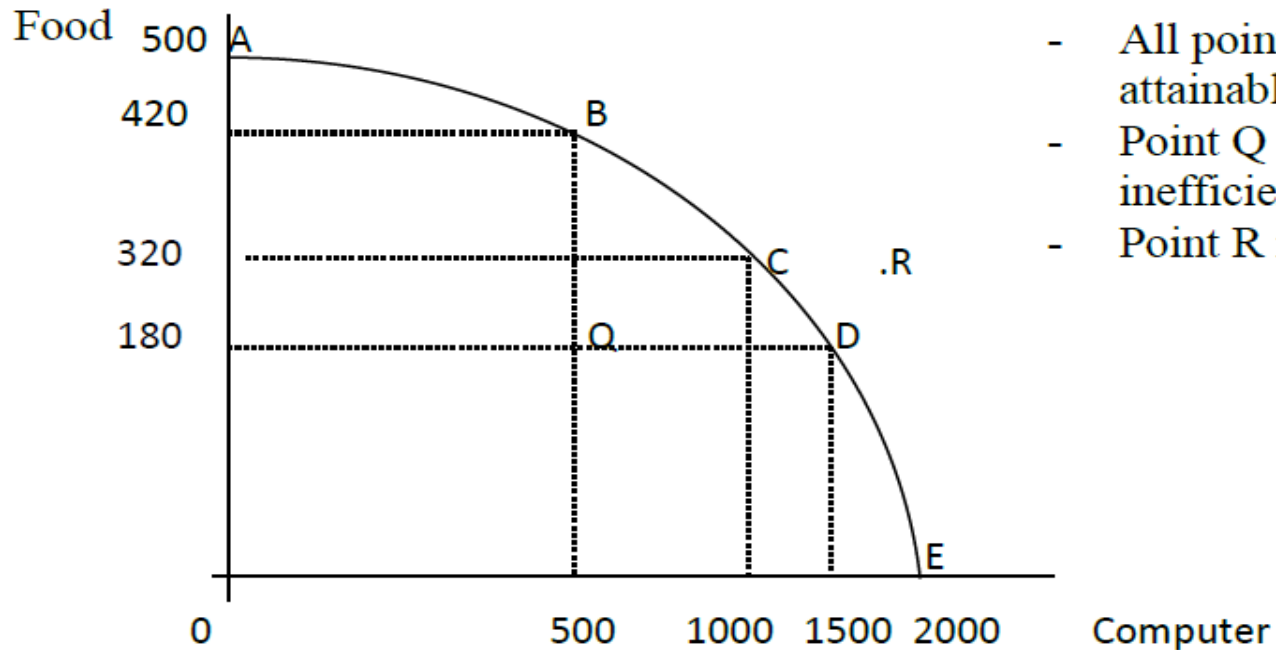
Examples

- Suppose a hypothetical economy produces food and computer given its limited resources and available technology

Types of products	Unit	Production alternatives				
		A	B	C	D	E
Food	metric tons	500	420	320	180	0
Computer	number	0	500	1000	1500	2000

Examples

The production possibility frontier (PPF)



- All points on the PPF are attainable and efficient
- Point Q is attainable but inefficient
- Point R is unattainable

The production possibility frontier (PPF)

The PPF describes four important concepts:

- **Scarcity:** the society cannot have unlimited amount of outputs even if it employs all of its resources and utilizes them in the best possible way.
- **Choice:** any movement along the curve indicates the change in choice.
- **Efficiency:** all points on the PPF are efficient
- **Opportunity cost:** when the economy produces on the PPF, production of more of one good requires sacrificing some of another product which is reflected by the downward sloping PPF.
 - **Law of increasing opportunity cost:** This law states that as we produce more and more of a product, the opportunity cost per unit of the additional output increases. This makes the shape of the PPF **concave** to the origin.

PPF & OC

- Calculating opportunity cost:

$$\text{Opportunity cost of a good} = \frac{\text{the amount of the next best alternative sacrificed}}{\text{the amount of the good gained}}$$

Say the economy is initially operating at **point B**,
What is the opportunity cost of producing one more unit of computer?

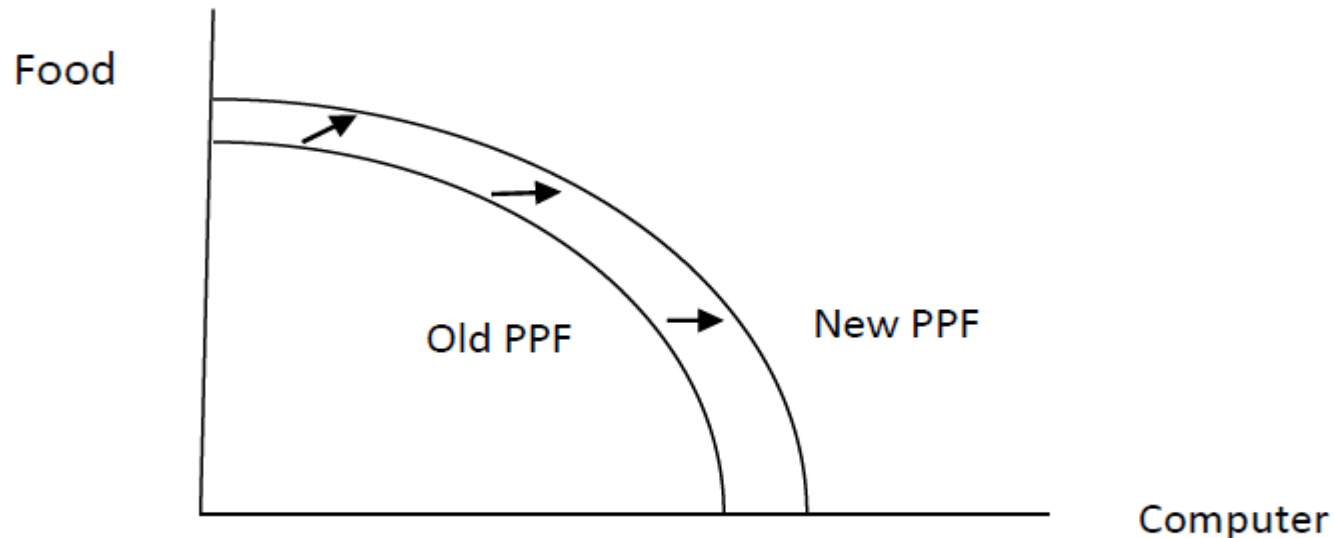
Solution: Moving from production alternative B to C we have:

$$OC = \frac{320-420}{1000-500} = \left| \frac{-100}{500} \right| = 0.2$$

The economy gives up 0.2 metric tons of food per computer.

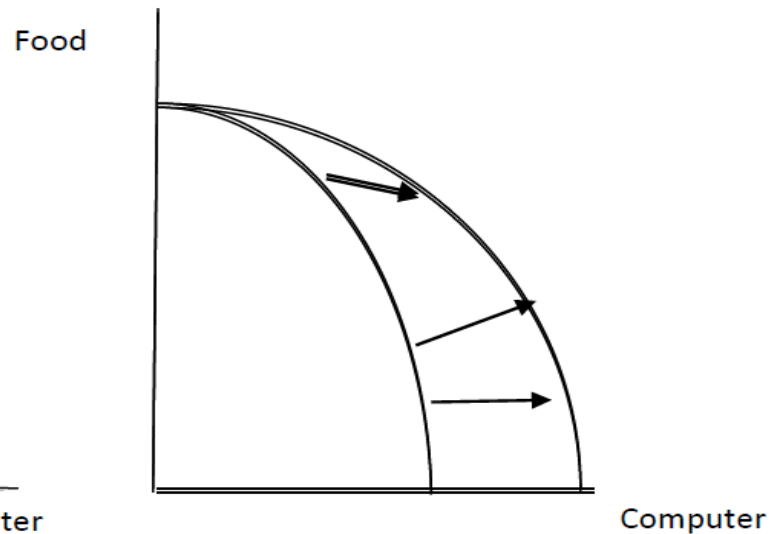
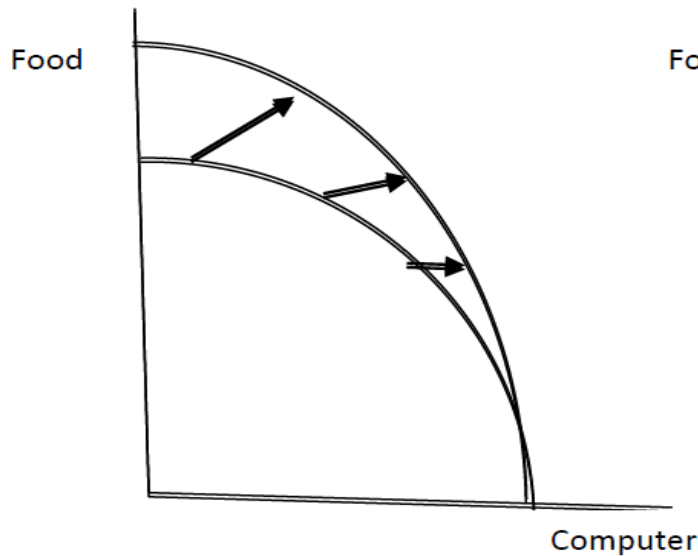
Economic Growth and the PPF

- Economic growth or an increase in the total output level occurs when one or both of the following conditions occur.
 - Increase in the quantity or/and quality of economic resources
 - Advances in technology
- Economic growth is represented by outward shift of the PPF.



Asymmetric growth and the PPF

- An economy can grow because of an increase in productivity in one sector of the economy. For example, an improvement in technology applied to either food or computer would be illustrated by a shift of the PPF along the Y-axis or X-axis.



Basic economic questions

- The fundamental economic problem leads to three important questions related to the allocation of resources.
- 1. What to Produce?** - the *problem of allocation of resources*.
 - 2. How to Produce?** - the *problem of choice of technique*.
 - 3. For Whom to Produce?** - the *problem of distribution of national product*.

What to Produce?

- Economies cannot produce everything, so they must decide **what to produce** and in **what quantities**.
- The economy must make choices such as
 - consumption goods versus capital goods,
 - civil goods versus military goods, and
 - necessity goods versus luxury goods.
- Generally, the final choice of any economy is a combination of the various types of goods but the exact nature of the combination depends upon the specific circumstances and objectives of the economy.

How to Produce?

- Once an economy has chosen what to produce , the economy must decide how to produce them.
- choosing between alternative methods or techniques of production.
 - For example, cotton cloth can be produced with hand looms, power looms, or automatic looms.
 - Similarly, wheat can be grown with primitive tools and manual labor, or with modern machinery and little labor.

How to Produce?

- Broadly speaking, the various techniques of production can be classified into two groups:
 1. labour-intensive techniques ,and
 2. capital-intensive techniques
- The choice between different techniques depends on the available **supplies** of different factors of production and their **relative prices**.
- Making **good choices** is essential for making the **best** possible use of limited resources to produce **maximum** amounts of goods and services.

For Whom to Produce?

- This problem is also known as the problem of distribution of national product.
- It relates to how a material product is to be distributed among the members of a society.
- The economy must decide, for example, whether to produce for the **benefit** of the **few rich people** or for **the large number of poor people**.
- An economy that wants to benefit the maximum number of persons would first try to produce the **necessities** of the whole population and then to proceed to the production of **luxury goods**.

Activity

In 2019, 28% of Japan's population was aged 65 years and over; by 2050, this figure is estimated at 38%. Total population is also projected to fall from 125m in 2019 to 100m in 2050. Suggest the likely effect of these changes on

- a) what to produce
- b) how to produce
- c) For whom to produce.

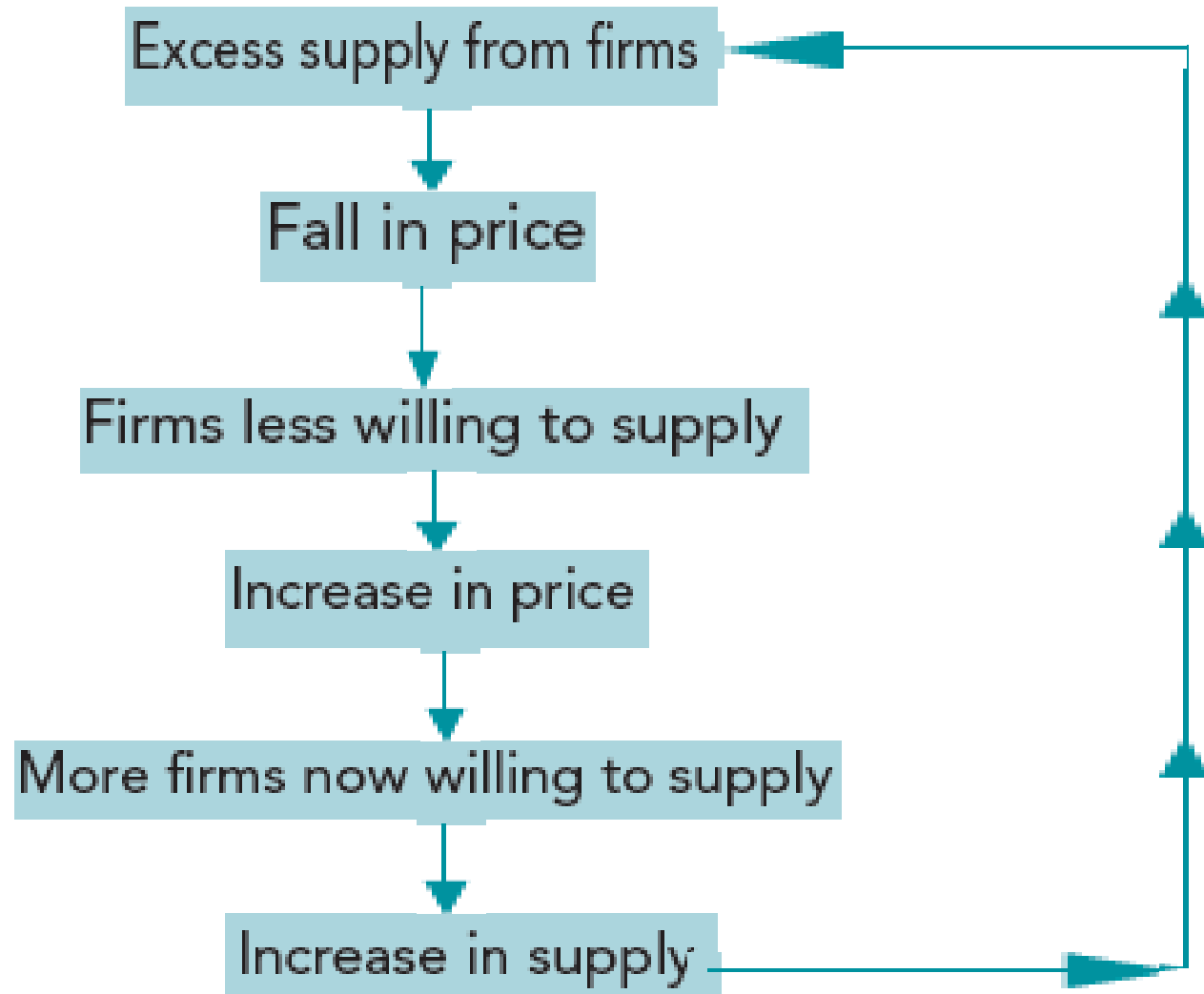
Economic systems

- The **way** a society tries to answer the above **fundamental questions** is summarized by a concept known as economic system.
- An **economic system** is a set of organizational and institutional arrangements established to answer the basic economic questions.
- **Three types** of economic system based on **resource allocation mechanisms / ownership of means of production**:
 - **Capitalism/Free market economy/laissez faire**
 - **Command/Socialistic/Planned economy**
 - **Mixed economy**

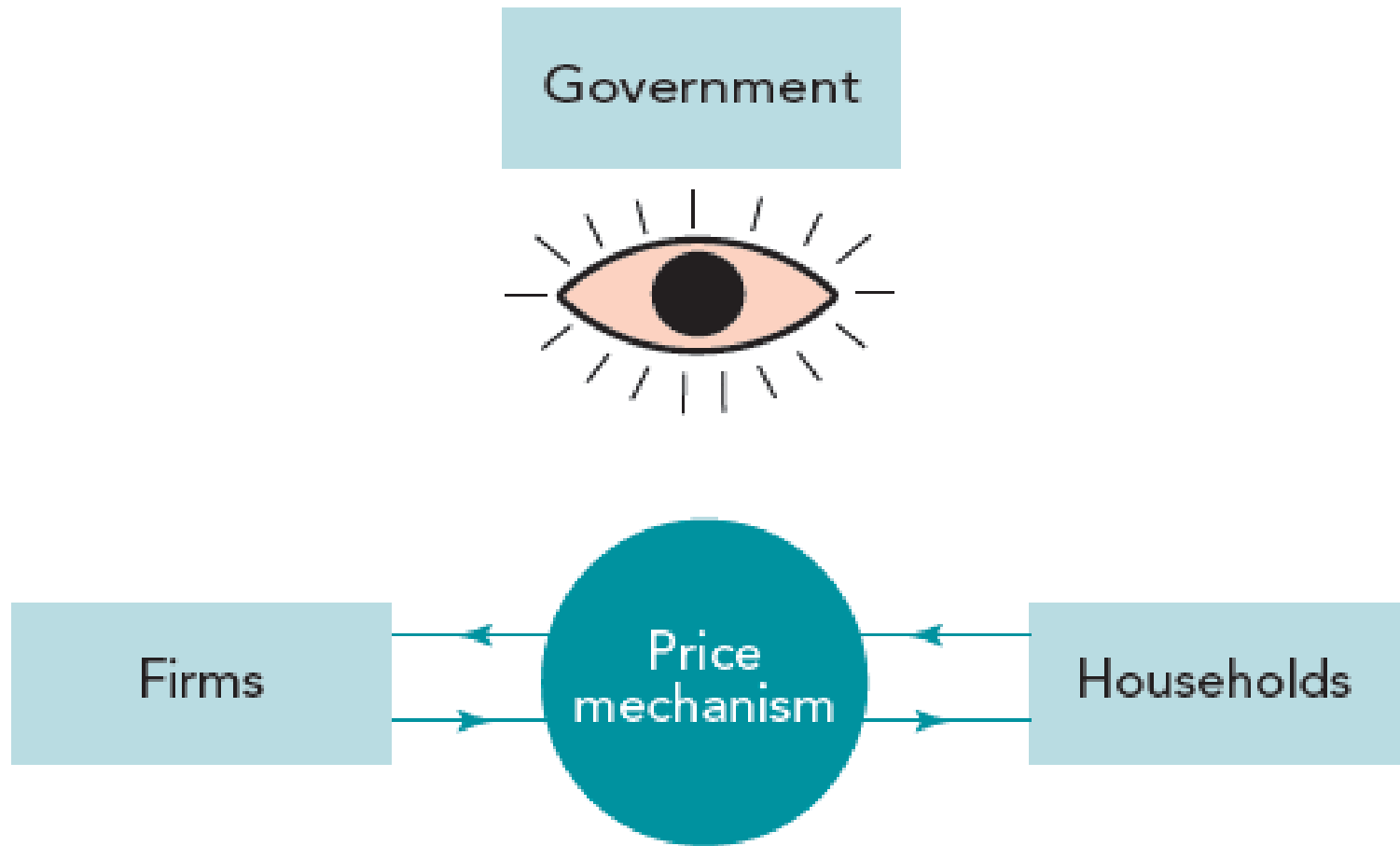
Capitalist economy

- In this economic system,
 - all means of production are privately owned,
 - and production takes place at the initiative of individual private entrepreneurs who work mainly for private profit.
 - Government intervention in the economy is minimal.
- resource allocation decisions are largely driven by the **market mechanism**, whereby individuals and firms take these decisions without government intervention.
- resources are allocated by the forces of **demand** and **supply** through the **price mechanism**

How the price mechanism works



Role of the government in a market economy



The market economy

Role of the government in a market economy

- In principle, the government should have no direct role in the workings of the market economy and should not interfere in the operation of the price mechanism.
- As long as the price mechanism is working efficiently, the government's role is to watch what is happening and only intervene when the price mechanism does not provide the best allocation of resources.
- When the price mechanism does not work efficiently, the market 'fails'.

Role of the government in a market economy

- Examples of market failure are
 - where the government provides goods such as **healthcare** which would be underprovided, **fire services** which would not be provided at all,
 - and where it seeks to **regulate situations** to **prevent firms** using their **power to control** the market for **excessive** gain.
- The **true market economy** is in certain respects an **ideal**.
- No actual economy operates as a pure market economy.
- **Hong Kong SAR** provides an interesting case of a near-perfect example of a market economy. All businesses are privately owned and Hong Kong's bus, rapid transit rail service (MTR) and road tunnels are operated by private companies.

Features of Capitalistic Economy

1. The right to private property
2. Freedom of choice by consumers
3. Productive activity is guided by the motive of profit-making and self-interest
4. Competition among economic agents prevail
5. All basic economic problems are solved through the market or price mechanism
6. Minor role of government
7. Inequalities of income
8. Existence of negative externalities

Capitalist Economy

Advantage

- Flexibility/adaptability
- Decentralized economic power
- Increase in per-capita income and standard of living
- Diversified types of consumer goods and services
- Growth of entrepreneurship
- Optimum utilization of productive resources
- High rate of capital formation

Disadvantage

- Merit goods are under-provided / demerit goods are overprovided. Eg. Junk food
- Unbalanced economic activity
- Inequality of income
- Exploitation of labor
- Negative externalities
- Excessive market power

Command economy

- Command economy is also known as **socialistic economy**.
- In this economic system,
 - The **economic institutions** that are engaged in production and distribution are **owned** and controlled by **the state**.,
 - the government has a **central role** in all decisions that are made.
 - The **choices** in terms of **what to produce**, **how to produce** and for **whom to produce** are all **centralized**.
 - **Unlike** the market economy, the **preferences of consumers** and **manufacturing organizations** in a planned economy are controlled **centrally**.

Main Features of Command Economy

1. **Collective ownership:** All means of production are owned by the society as a whole, and there is no right to private property.
2. **Central economic planning:** Planning for resource allocation is performed by the controlling authority according to given socio-economic goals.
3. **Strong government role:** Government has complete control over all economic activities.
4. **Maximum social welfare:** Command economy aims at maximizing social welfare and does not allow the exploitation of labor.
5. **Relative equality of incomes:** Private property does not exist in a command economy, the profit motive is absent, and there are no opportunities for accumulation of wealth. All these factors lead to greater equality in income distribution, in comparison with capitalism.

Command economy

Advantage

- Absence of wasteful competition
- Balanced economic growth
- Elimination of private monopolies and inequalities
- Controls labor exploitation and negative externalities

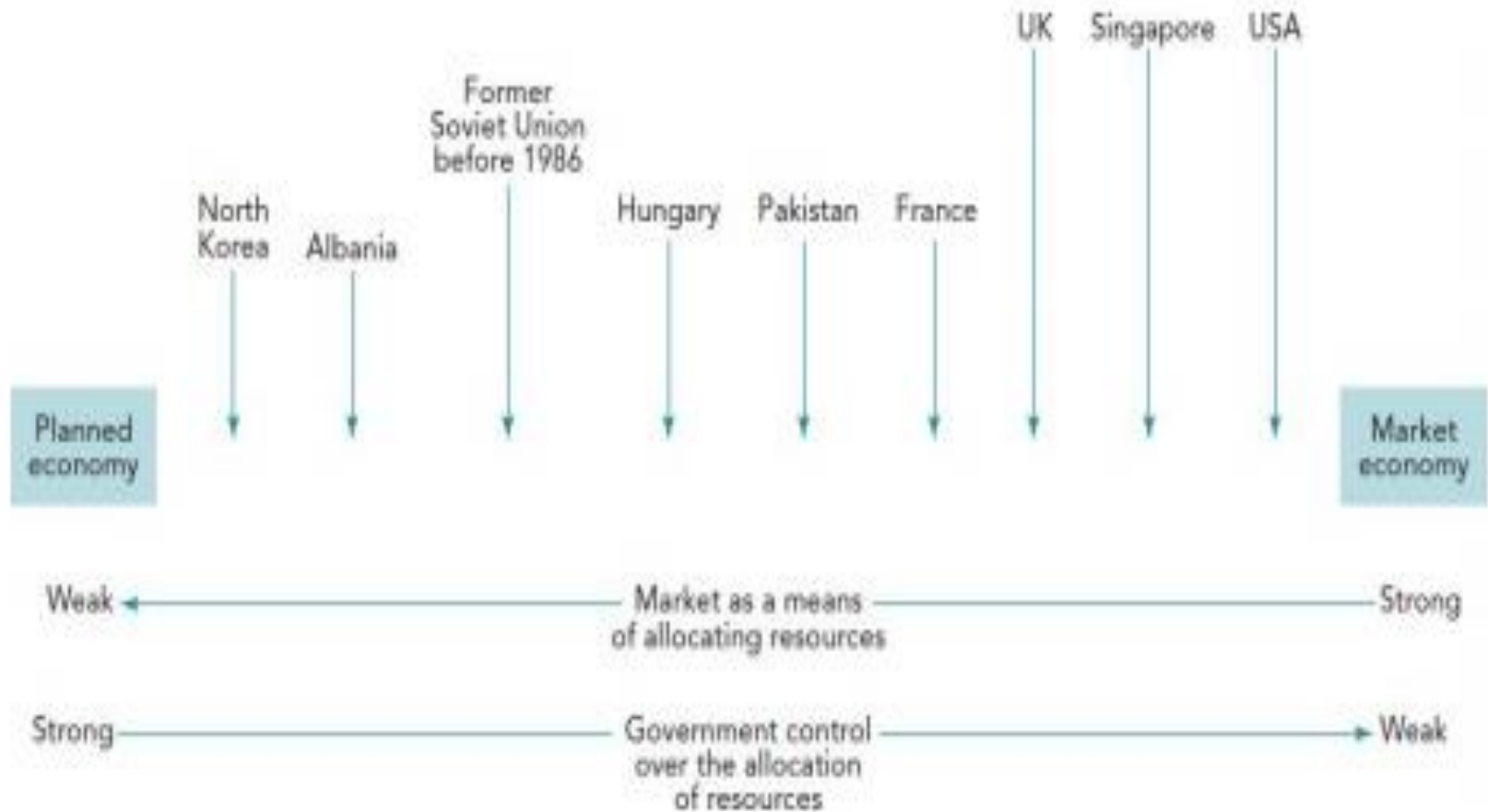
Disadvantage

- Absence of automatic price determination
- Absence of incentives for hard work and efficiency
- Lack of economic freedom
- Red-tape

Mixed economy

- Incorporates some features of both capitalist & command economy
- Co-existence of public and private sectors
- Economic welfare where public sector tries to remove regional imbalances, provides large employment opportunities and seeks economic welfare through its price policy
- Economic planning to achieve co-ordinated rapid economic development, making use of both the private and the public sector.
- Price mechanism

Mixed market systems



Mixed economy

Advantages

- Private property, profit motive and price mechanism
- Adequate freedom
- Rapid and planned economic development
- Social welfare and fewer economic inequalities

Disadvantages

- Ineffectiveness and inefficiency
- Economic fluctuations
- Corruption and black markets

Discussion questions

- What are examples of the most well-known countries with market/command/mixed economies?
- What are? How have different societies that you know about attempted to answer the three basic economic questions?

Decision making units and the circular flow model

□ Decision making units

Agents	Who is?	Decisions /roles
1.Household	One person or more who live under one roof and make joint financial decisions	- Selling of their resources - Buying of goods and services
2. Firm	Production unit that uses economic resources to produce goods and services	- Buying of economic resource - Selling of their products
3. Government	An organization that has legal and political power to control or influence households, firms and markets	- Provide public goods and services - Create enabling business environment - Regulation, law and order

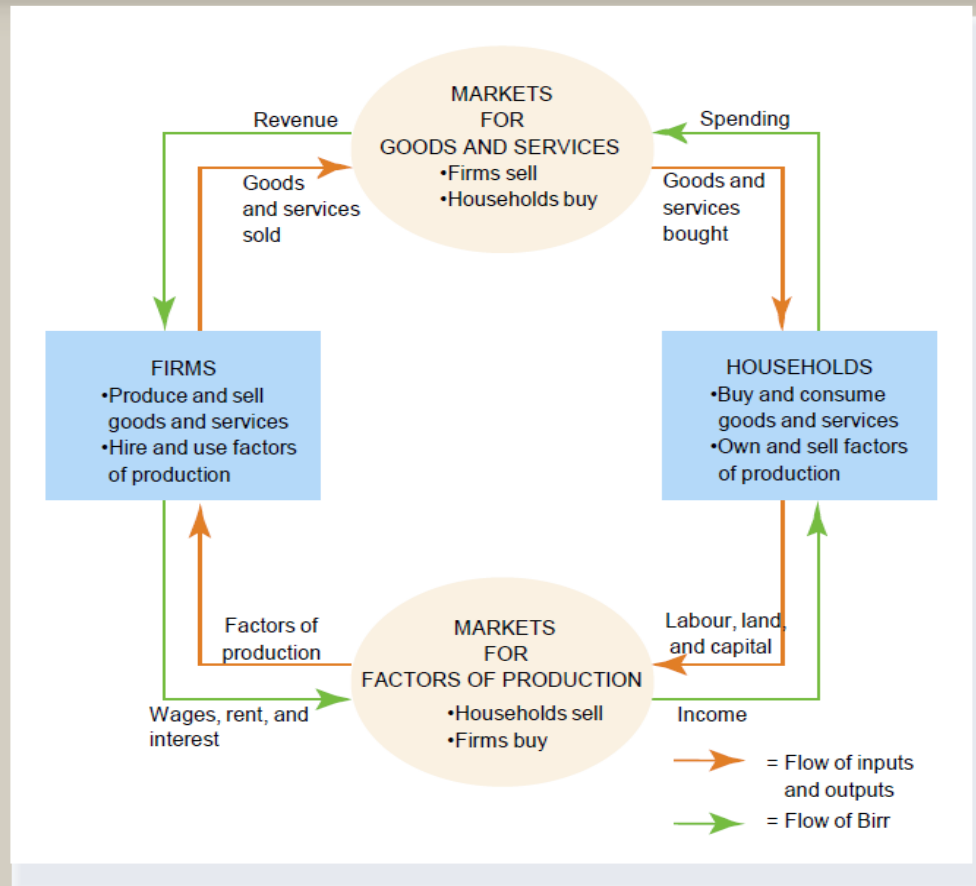
Decision making units and the circular flow model

□ Markets

- **Product market:** it is a market where goods and services are transacted/ exchanged. That is, a market where households and governments buy goods and services from business firms.
- **Factor market (input market):** it is a market where economic units transact/exchange factors of production (inputs). In this market, owners of resources (households) sell their resources to business firms and governments.

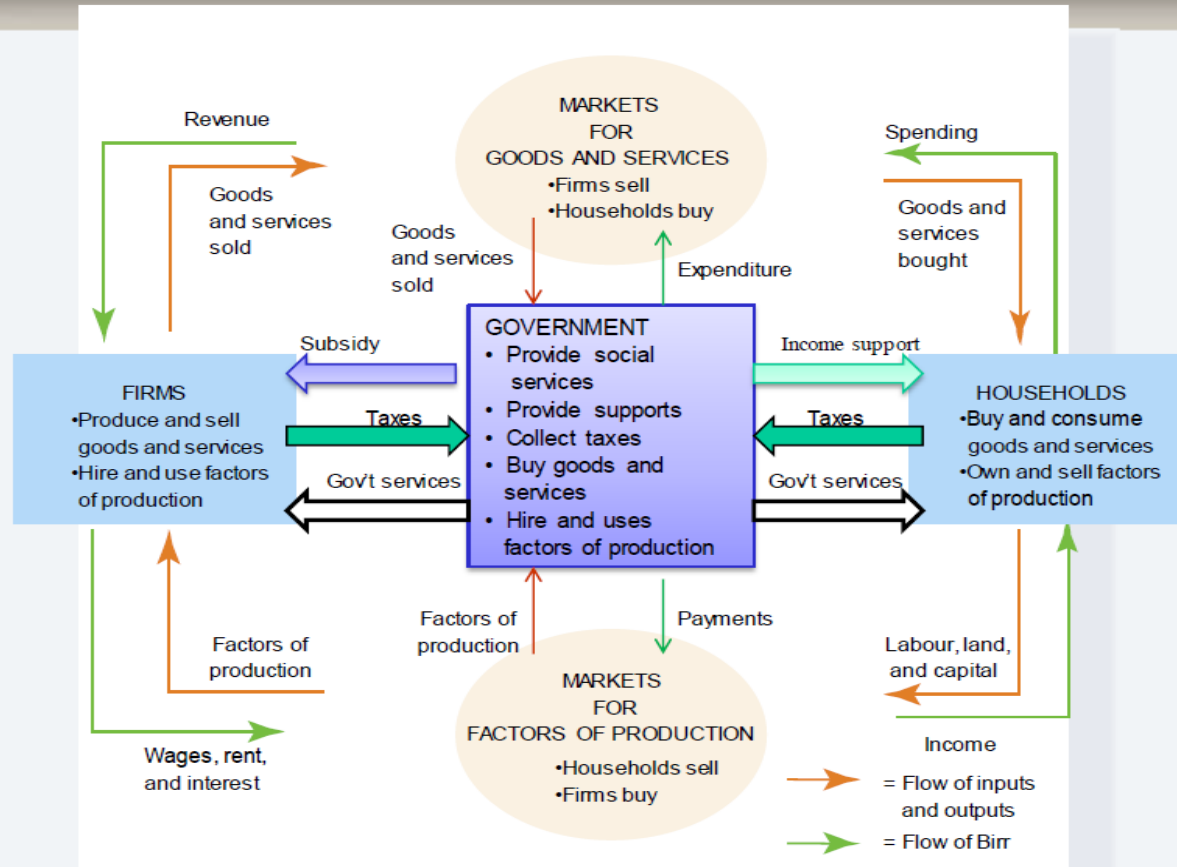
Circular-flow of the economy

A Two sector model



Circular-flow of the economy

A Three sector model



- EXERCISES (multiple choice and essay)